

AP Precalculus Unit 2 Wkst B
Graphing Secant and Cosecant

Name: _____

Graph one period of each of the following $g(x)$ functions. Be sure to include the midline and any asymptotes on your graph. Graphs **MUST** be done on graph paper.

Then, describe the transformation (translation/shift, dilation/stretch-shrink, reflection) of $g(x)$ compared to $f(x)$. Be specific. Make sure to tell direction and amount of the transformation.

Then, state the domain and the range of the $g(x)$ function. And use interval notation to state the interval(s) where the function values are increasing, while the rate of change is decreasing.

1. $g(x) = \csc \pi x$ $f(x) = \csc x$

2. $g(x) = \sec \pi x - 1$ $f(x) = \sec x$

3. $g(x) = -2 \sec 4x + 2$ $f(x) = \sec x$

4. $g(x) = -3 \csc (\pi x + 3\pi) - 4$ $f(x) = \csc x$

5. $g(x) = 5\csc(4x - \pi)$ just graph it